Introduction to Quantum Computing

1. What is Quantum Computing

Quantum computing is a way of processing information using the principles of quantum mechanics. Instead of bits, it uses qubits, which allows certain complex problems to be approached more efficiently than with classical computers.

- Uses qubits instead of bits
- Built for specialised, complex problems, not everyday tasks
- Does not replace classical computers

Analogy: A Very Large Puzzle

A classical computer checks possibilities one by one. A 20-piece puzzle is easy, but a puzzle with billions of combinations quickly becomes very slow to solve this way. Quantum computing offers a different approach for problems of this scale.

Where It Is Used

Industry	Problem
Healthcare	Drug discovery
Chemistry	Molecular simulation
Finance	Portfolio optimisation
Logistics	Route optimisation
Artificial Intelligence	Complex model training
Cybersecurity	Advanced cryptography research

2. Classical Bit vs Quantum Qubit

A bit stores information as either 0 or 1, one value at a time. A qubit can represent 0, 1, or a combination of both until it is measured. This property is called superposition.

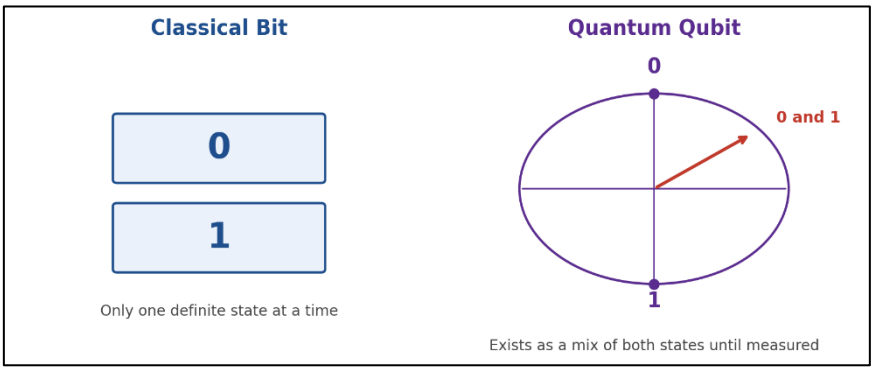
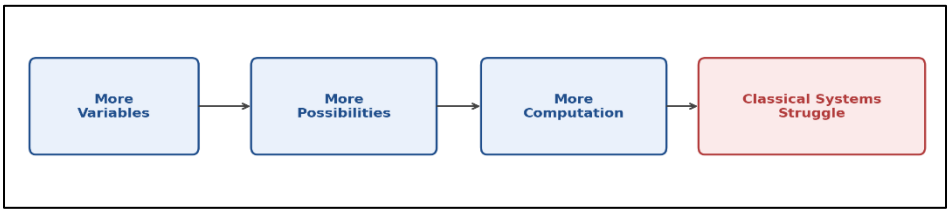


Figure 1: A classical bit holds one fixed state; a qubit holds a mix of states until measured.

3. Why Quantum Computing Exists

As real-world problems grow larger, the number of variables and calculations grows rapidly, and classical systems begin to struggle to keep up.



Example: Drug Discovery

To understand how a drug molecule behaves, scientists must simulate millions of interactions between atoms and electrons. More atoms lead to more interactions, more calculations, and exponential growth in the work required. This is one of the reasons quantum computers are of interest, since nature itself follows quantum mechanics.

4. Core Principles of Quantum Computing

4.1 Qubits

The qubit is the fundamental unit of quantum information. More qubits allow many more possible states to be represented at once. With n qubits, there are 2 to the power n possible states.

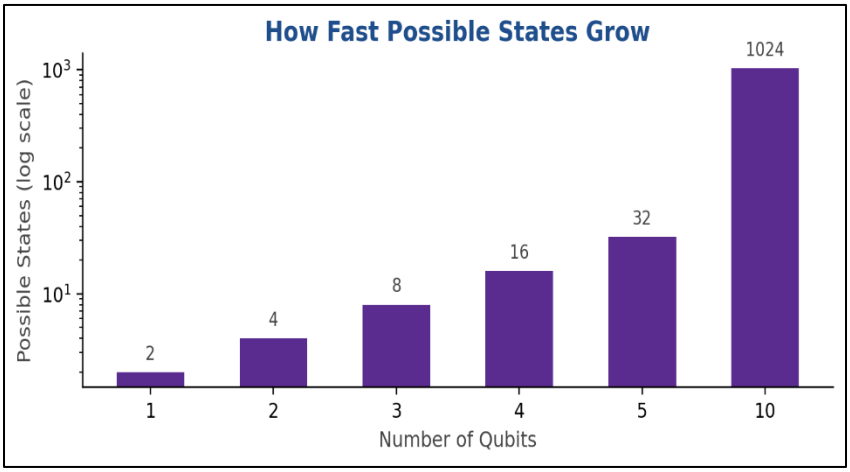


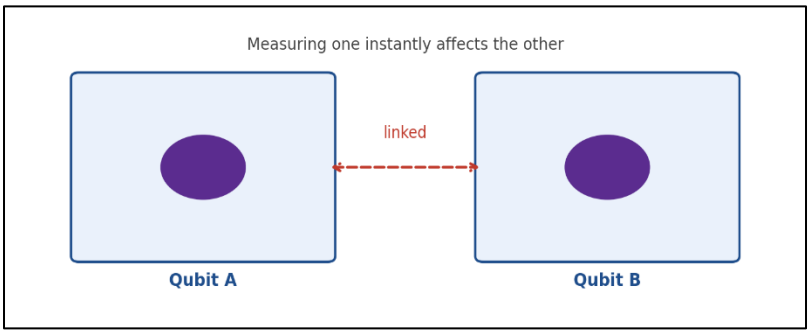
Figure 2: Possible states grow exponentially with each added qubit.

4.2 Superposition

Superposition allows a qubit to exist in multiple possible states at the same time, before it is measured. Once measured, it settles into a single definite outcome, similar to a spinning coin that is both heads and tails until it lands.

4.3 Entanglement

When two qubits become entangled, they are linked so that measuring or changing one instantly affects information about the other, even if they are physically separated. This effect enables many powerful quantum algorithms.



4.4 Interference

Quantum algorithms use interference to amplify the probability of correct answers and suppress incorrect ones, increasing the chance that a measurement returns the desired result.

4.5 Decoherence

Quantum information is fragile. Environmental noise can disturb a qubit, causing information loss. This process, called decoherence, is one of the biggest engineering challenges in building reliable quantum computers.

5. Classical vs Quantum Computing

Feature	Classical	Quantum
Unit	Bit	Qubit
States	0 or 1	0 and 1
Processing	Deterministic	Probabilistic
Noise sensitivity	Low	High
Everyday tasks	Excellent	Poor
Optimisation problems	Limited	Potentially strong

6. IBM Quantum and Qiskit

IBM provides real quantum hardware, cloud access, learning resources, and development tools. Qiskit is IBM's open-source quantum software development kit, often described as Python for quantum computing. It allows developers to build circuits, simulate programs, run jobs on real quantum computers, and analyse results.

IBM Quantum Stack

- Developer writes code
- Qiskit builds and manages quantum circuits
- IBM Runtime executes the workload
- Quantum hardware processes the job

Results are returned for analysis

Summary: What to Remember

- Quantum computing uses qubits, which can hold 0, 1, or both at once through superposition
- It is built for specific complex problems, not to replace everyday computing
- More qubits mean exponentially more possible states (2 to the power n)
- Entanglement links qubits so that measuring one affects the other
- Interference boosts correct answers and reduces incorrect ones
- Decoherence from noise is the main challenge to overcome
- Qiskit and IBM Quantum provide practical tools to start learning and building